

Drift-diffusion problems with Gauss–Fermi statistics and field-dependent mobility for organic semiconductor devices

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We consider a drift-diffusion model for organic semiconductor devices including Gauss–Fermi statistics and application-specific mobility functions. The charge transport in organic materials is realized by hopping of carriers between adjacent energetic sites and is described by complicated mobility laws with a strong nonlinear dependence on temperature, carrier densities and the electric field strength. We report on the existence of weak solutions to the stationary problem as well as global weak solutions to the instationary problem. Moreover, L^∞ bounds for the solutions are established.

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1 Drift-diffusion modeling of organic semiconductors

Organic semiconductor devices are carbon-based thin-film structures formed by small molecules or polymers. In the literature (e.g. [4, 5] and references therein), organic materials are modeled at different scales, ranging from density functional theory for molecules, master equation approaches for carrier dynamics in homogeneous materials, to drift-diffusion equations. The electronic transport in such materials is realized by hopping of electrons between discrete energy levels of adjacent molecules. The energy levels are Gaussian-distributed, centered at $E_{n,p}$ with disorder parameter $\sigma_{n,p}$ such that the carrier densities n, p in the presence of an electric field $-\nabla\psi$ are related to the quasi-Fermi potentials $\varphi_{n,p}$ by Gauss–Fermi integrals, i.e.

$$n = N_{n0} \mathcal{G}_n \left(\frac{q(\psi - \varphi_n) - E_n}{k_B T} \right) = \frac{N_{n0}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp \left(-\frac{x^2}{2} \right) \frac{1}{\exp \left(\frac{\sigma_n x}{k_B T} - \frac{q(\psi - \varphi_n) - E_n}{k_B T} \right) + 1} dx \quad (1)$$

with total density of transport states N_{n0} , elementary charge q , temperature T and Boltzmann's constant k_B . Figure 1 shows this statistical relation in comparison to Boltzmann and Fermi–Dirac statistics. In contrast to classical semiconductors with Boltzmann statistics, this leads to a generalized Einstein relation and results in a nonlinear diffusion enhancement in the relation between drift and diffusion current. The mobility functions $\mu_{n,p}$ for organic semiconductor materials with Gaussian disorder are increasing with respect to temperature, carrier density, and electric field strength. These dependencies are obtained from numerically solving the master equation for hopping transport in a disordered energy landscape, see [1, 5]. For a drift-diffusion-type description of the charge transport we use an adapted van Roosbroeck system consisting of continuity equations for the electron and hole densities n and p which are self-consistently coupled to a Poisson equation for the electrostatic potential ψ . After rescaling of potentials and energies the system takes the form

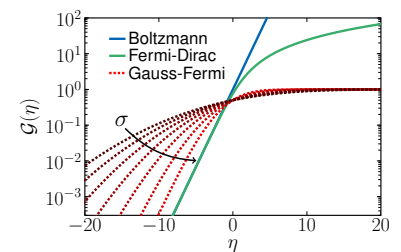


Fig. 1: Statistical relation compared to Boltzmann and Fermi–Dirac statistics

$$\begin{aligned} -\nabla \cdot (\varepsilon \nabla \psi) &= C - n + p, \\ \partial_t n - \nabla \cdot j_n &= -R, \quad j_n = -n \mu_n(T, n, |\nabla \psi|) \nabla \varphi_n, \quad n = e_n(\psi - \varphi_n) := N_{n0} \mathcal{G}_n(\psi - \varphi_n - E_n), \\ \partial_t p + \nabla \cdot j_p &= -R, \quad j_p = -p \mu_p(T, p, |\nabla \psi|) \nabla \varphi_p, \quad p = e_p(\varphi_p - \psi) := N_{p0} \mathcal{G}_p(\varphi_p - \psi + E_p). \end{aligned} \quad (2)$$

C denotes a fixed doping density, the driving forces in the carrier fluxes are the gradients of the (scaled) quasi-Fermi potentials $\varphi_{n,p}$, and the generation-recombination term has the form $R = r(\cdot, n, p)(1 - e^{\varphi_n - \varphi_p}) = r_0(\cdot, n, p)np(1 - e^{\varphi_n - \varphi_p})$. The system is completed by initial conditions n^0 and p^0 and mixed boundary conditions, prescribing Dirichlet values $\varphi_{n,p}^D, \psi^D$ at the contacts Γ_D and homogeneous Neumann boundary conditions on the remaining part of $\partial\Omega \setminus \Gamma_D$.

2 The stationary problem

The existence and boundedness of stationary weak solutions for problem (2) in any spatial dimension can be verified by means of Schauder's fixed point theorem. We consider the non-empty, convex, closed, precompact set

$$\mathcal{X} := \{(\varphi_n, \varphi_p) \in L^2(\Omega)^2 : \|\varphi_i\|_{L^\infty} \leq \|\varphi_{n,p}^D\|_{L^\infty}, \|\varphi_i\|_{H^1} \leq c^*, i = n, p\},$$

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where c^* is defined later, and introduce the fixed point map $Q : \mathcal{X} \rightarrow \mathcal{X}$ as follows: For arbitrary $(\tilde{\varphi}_n, \tilde{\varphi}_p) \in \mathcal{X}$ there is a unique solution ψ to the nonlinear Poisson equation

$$-\nabla \cdot (\varepsilon \nabla \psi) = C - e_n(\psi - \tilde{\varphi}_n) + e_p(-(\psi - \tilde{\varphi}_p))$$

(since the corresponding operator is strongly monotone and Lipschitz continuous). Then we derive the bounds $\|\psi\|_{L^\infty} \leq c$, $\tilde{n} := e_n(\psi - \tilde{\varphi}_n)$, $\tilde{p} := e_p(-(\psi - \tilde{\varphi}_p)) \in [c_1, c_2]$ and determine (φ_n, φ_p) as unique solution to

$$\nabla \cdot (\tilde{n} \mu_n(\tilde{n}, |\nabla \psi|) \nabla \varphi_n) + r(\tilde{n}, \tilde{p})(1 - e^{\varphi_n - \varphi_p}) = 0, \quad -\nabla \cdot (\tilde{p} \mu_p(\tilde{p}, |\nabla \psi|) \nabla \varphi_p) + r(\tilde{n}, \tilde{p})(1 - e^{\varphi_n - \varphi_p}) = 0$$

and fix c^* as a uniform bound for $\|\varphi_i\|_{H^1} \leq c^*$, $i = n, p$. Finally, the continuity of Q is verified, for details see [2]. As for classical semiconductor devices, uniqueness of the solution cannot be expected.

3 The instationary problem

We use the weak formulation of problem (2) in the form $u' + A(v) = 0$, $u = E(v)$, $u(0) = u^0$ with the variables $v = (v_0, v_n, v_p) := (\psi, \psi - \varphi_n, \varphi_p - \psi)$ and $u = (u_0, u_n, u_p) := (u_0, n, p)$, $\langle u_0, w \rangle = \int_\Omega (p - n)w \, dx \quad \forall w \in H_0^1(\Omega \cup \Gamma_N)$ and

$$\begin{aligned} \langle E_0 v_0, \bar{v}_0 \rangle &:= \int_\Omega (\varepsilon \nabla v_0 \cdot \nabla \bar{v}_0 - C \bar{v}_0) \, dx, \quad E(v) := (E_0 v_0, e_n(v_n), e_p(v_p)), \\ \langle A(v), \bar{v} \rangle &:= \int_\Omega e_n(v_n) \mu_n(e_n(v_n), |\nabla v_0|) \nabla(v_n - v_0) \cdot \nabla(\bar{v}_n - \bar{v}_0) + e_p(v_p) \mu_p(e_p(v_p), |\nabla v_0|) \nabla(v_p + v_0) \cdot \nabla(\bar{v}_p + \bar{v}_0) \, dx \\ &\quad + \int_\Omega r(e_n(v_n), e_p(v_p))(1 - e^{-v_n - v_p})(\bar{v}_n + \bar{v}_p) \, dx \quad \forall \bar{v} \in H_0^1(\Omega \cup \Gamma_N)^3. \end{aligned}$$

The Dirichlet boundary conditions at Γ_D are formulated via $v|_{\Gamma_D} = v^D$. In two spatial dimensions the existence of global weak solutions of the instationary problem is obtained by considering a problem with regularized state equations and reaction term $u' + A_M(u, v) = 0$, $u = E_M(v)$, $u(0) = u^0$. Its solvability is shown by time discretization (where arguments u_i and $|\nabla \psi|$ in the mobility function and in the reaction rate coefficient are taken from the previous time step) and passage to the time-continuous limit. Positive lower a priori estimates for the densities of solutions for the regularized problem that are independent of the regularization level M ensure the existence of solutions to the original problem, see [3]. To derive positive lower bounds for the charge carrier densities u_i for the regularized as well as the original problem, we use the test functions

$$-m e^{mt} \frac{z^{m-1}}{u_i}, \quad z := (\ln u_i + K^*)^-, \quad K^* := \max \left\{ \max_{i=n,p} \|\ln e_i(v_i^D)\|_{L^\infty}, \max_{i=n,p} \|(\ln u_i^0)^-\|_{L^\infty} \right\},$$

$m = 2^k$, $k \in \mathbb{N}$, and apply a Moser iteration procedure. For problems with Gauss–Fermi statistics, the charge carrier densities are automatically bounded by the total density of transport states N_{n0} , see Fig. 1. On the one hand, due to the special properties of the statistical relation (1), powers of u_i and also of v_i are not suited to find global upper bounds strictly below this total density of transport states and to establish the needed global upper bounds for the chemical potentials v_i . On the other hand, since it is not a priori clear that the functions

$$m e^{mt} \frac{[(e^{v_i} - K)^+]^{m-1} e^{v_i}}{e_i'(v_i)}, \quad K := \max \left\{ \max_{i=n,p} e^{\|v_i^D\|_{L^\infty}}, \max_{i=n,p} e^{\|e_i^{-1}(u_i^0)\|_{L^\infty}}, \max \{e^{\sigma_n + E_n}, e^{\sigma_p - E_p}\} \right\}, \quad (3)$$

are admissible test functions in $L_{\text{loc}}^2(\mathbb{R}_+, H_0^1(\Omega \cup \Gamma_N))$, we approximate them by substituting v_i in (3) with $v_L := \min(v_i, L)$, apply in this setting the Moser iteration technique, and consider finally the limit $L \rightarrow \infty$, see [3]

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